

MEASURING STATE ALIASING IN NEURAL MULTI-FRAGMENT ROUTING

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ABSTRACT

Flexible construction order changes the information a neural routing decoder needs. We study a specific interface mismatch: a route decoder reused to join directed path fragments can observe the same static embeddings, selected-source context, and feasibility mask in states requiring different optimal endpoints. We quantify this ambiguity using exact conditional completion costs and the minimum regret of a policy constrained to share its action across indistinguishable states. A seven-vertex witness admits a numerically certified positive regret; controlled paired forests derived from 1,536 larger Euclidean instances exhibit conflicting optimal decisions in 20.7–29.7% of pairs across six size/distribution conditions. We distinguish this constructed stress test from actual policy occupancy and separate representational limits from optimization failure. Controlled neural prediction and official-backbone adaptation experiments test whether making component connectivity observable improves decisions and complete tours. The study treats paired-endpoint representations and state-abstraction principles as existing ideas, and evaluates their role in a concrete decoder transfer failure. Conditional information gains do not, by themselves, establish an advantage for learned construction order or a stronger routing solver.

1 INTRODUCTION

Neural routing models commonly construct one path by repeatedly selecting its next vertex. More flexible construction can instead maintain several disjoint paths and choose which tail to extend. This appears to be a small change to the action space: select a source, call a pretrained endpoint decoder, and mask illegal merges. However, a mask answers which actions are feasible, whereas the value of an action also depends on what those actions leave for the future. In a path forest, entering a target head commits the solution to traversing its entire component and leaving through its tail. Two forests can expose the same legal heads while connecting those heads to different exits.

This paper investigates that discrepancy in adapters around the Attention Model (AM; Kool et al. 2019) and ICAM (Zhou et al., 2026). The claim concerns the observation supplied by our forest adapters, not the correctness of either native solver. In an archived limited-budget adaptation study, learned source selection improves on trained random selection but remains worse than route continuation. That negative result does not identify its cause: insufficient training, incompatible representations, and an unhelpful action factorization remain distinct explanations. We therefore ask a narrower, testable question: *does the endpoint interface merge states that require different optimal decisions?*

We answer the representational question through exact completion, rather than teacher likelihood or a heuristic proxy. For an observation class, we compare the best state-specific actions with the best action shared across its members. Their difference is a standard conditional Bayes-risk gap. Its use here makes the missing information measurable and exposes a subtlety in evaluating counterexamples: the best shared action may be neither state’s best action. We then test information restoration with equal-architecture neural predictors and controlled residual adapters. Conditional prediction and complete-tour evaluation are reported separately.

Our contribution is an empirical and implementation-level audit: an exact diagnostic and certified witness, reproducible observation-matched state families, and controlled tests of the proposed explanation. Neither paired endpoint encoding nor path contraction is introduced as a new representation; DRHG already uses both endpoint coordinates for retained fragments (Li et al., 2025). The resulting

evidence must also be interpreted locally. A source selector can avoid ambiguous states or communicate information through its source choice. A lower bound for a fixed-source endpoint policy therefore does not bound the best jointly learned solver.

2 DIRECTED FORESTS AND ENDPOINT OBSERVATIONS

Let $X = (x_1, \dots, x_n)$ be Euclidean coordinates with distance $d(u, v)$. A partial solution F consists of m vertex-disjoint directed paths, including singletons. Its components have heads h_i and tails t_i . Existing directed edges are fixed: no path reversal, splitting, or reassignment is allowed. With $m > 1$, the action (i, j) joins t_i to h_j for $i \neq j$; the last component is closed into a tour. We assume a complete graph without capacity, time windows, or other side constraints.

Given a selected source component s , define $Q_F(s, a)$ as the minimum final tour cost among completions of F containing $t_s \rightarrow a$. The one-decision regret is

$$r_F(s, a) = Q_F(s, a) - \min_b Q_F(s, b). \quad (1)$$

This uses optimal continuation after one decision, not the rollout of an imperfect downstream model. Fixed internal path costs cancel from regret.

The audited endpoint observation is

$$z = \phi(X, F, s) = (E(X), h_s, t_s, M_{F,s}), \quad (2)$$

where $E(X)$ is a static node encoding and $M_{F,s}$ masks illegal heads. The decoder does not receive all head–tail correspondences, a recurrent construction history, or the source selector’s hidden state. Even an arbitrarily powerful static encoder cannot recover a dynamic correspondence absent from its input. This statement would not apply to an encoder recomputed from the forest. The restriction to the selected source’s mask row matters: the full matrix over all open tails and heads identifies each tail’s own component head as its unique forbidden open head, and therefore reveals their pairing.

Proposition 1 (Conditional observation regret). *For a fixed observation z , let μ_z be a distribution over forests with this observation and common finite action set A_z . The smallest expected regret among policies depending only on z and independent private randomness is*

$$\Delta(z) = \min_{a \in A_z} \mathbb{E}_{\mu_z} Q_F(s, a) - \mathbb{E}_{\mu_z} \min_{b \in A_z} Q_F(s, b) = \min_{a \in A_z} \mathbb{E}_{\mu_z} r_F(s, a). \quad (3)$$

For finite support with positive probability on each state, $\Delta(z) = 0$ exactly when all statewise optimal-action sets have a common member.

Proof. The expected regret of a randomized policy $p(a \mid z)$ is the convex combination $\sum_a p(a \mid z) \mathbb{E}_{\mu_z} r_F(s, a)$. Minimizing it selects a smallest coefficient. Every coefficient is nonnegative, and a coefficient vanishes exactly when its action has zero regret in every positive-mass state. $\square$

Equation (3) is a finite-action value-of-information identity, not a new general state-abstraction theorem (Li et al., 2006). It becomes an operational diagnostic when Q can be computed exactly. A two-state statistic with access to a sampled pair’s identity can be an optimistic lower bound for a less-informed decoder, rather than its full distributional Bayes risk. We distinguish these cases in our experiments.

Proposition 2 (Sufficient component state). *For the directed-forest problem above, all future actions, transition costs, and optimal continuation values are determined by the set of paired endpoints $\{(h_i, t_i)\}_{i=1}^m$ and the distance function. The remaining optimization is a directed TSP over components with cost $W_{ij} = d(t_i, h_j)$.*

Every completion traverses each fixed path contiguously and orders the paths cyclically. Conversely, every such order is feasible in a complete graph. Merging (i, j) replaces their pairs by (h_i, t_j) and incurs $d(t_i, h_j)$, which proves the transition claim. The fixed internal cost is needed only to recover the total objective. This standard contraction explains the repair; it does not prove that a particular neural pooling operation preserves all information. The contracted matrix is generally asymmetric even for symmetric Euclidean distances. Component means and sizes can aid approximation but are not necessary for the exact unconstrained continuation problem.

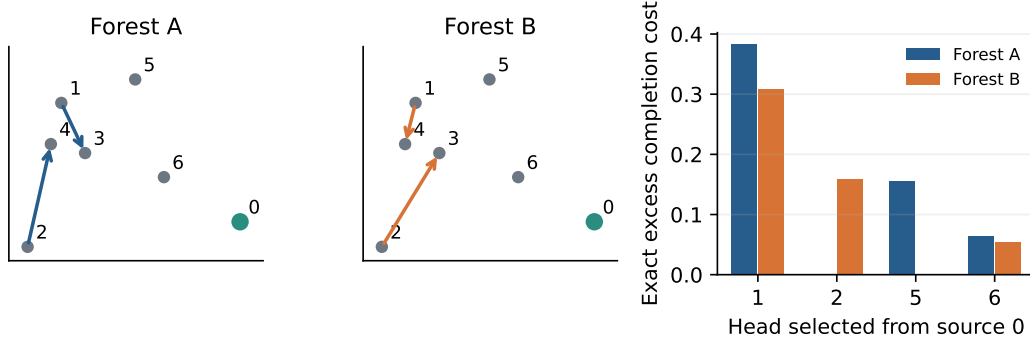


Figure 1: The selected source (green, vertex 0), legal heads, and static coordinates are identical. Fixed directed edges (left) change the optimal conditional completion costs (right). Head 6 is the best compromise for a decoder unable to distinguish the forests.

For $m \geq 2$, there are $(m-1)!$ directed completion orders and $(m-2)!$ completions after any legal merge. Every source has $m-1$ legal heads. Source choice alone therefore does not prune the reachable tour set or select a smaller legal domain. It may still change learning difficulty and the quality obtained by an imperfect endpoint policy. A partial-permutation or stabilizer-coset description does not alter these counts or confer an additional search guarantee.

3 EXACT DIAGNOSTICS AND CONTROLLED LEARNING

3.1 A CERTIFIED WITNESS

Figure 1 compares $F_A = \{1 \rightarrow 3, 2 \rightarrow 4\}$ with $F_B = \{1 \rightarrow 4, 2 \rightarrow 3\}$ on the same seven vertices and source 0. Both forests expose heads $\{1, 2, 5, 6\}$ and the same source context. The audited AM and ICAM endpoint vectors are exactly identical across the pair. The optimal heads are 2 and 5, respectively. However, the best *shared* action is head 6: the regrets are 0.06336798 and 0.05438468, giving $\Delta = 0.05887633055$. Using only the two statewise optimal actions would overestimate the bound.

An independent enumeration of all $6! = 720$ anchored tours reproduces the 24 legal completions of each forest. Interval arithmetic on the recorded decimal coordinates certifies $0.058876330550917 \leq \Delta \leq 0.058876330550931$. The certificate concerns that specified decimal instance; it makes no distributional or full-policy claim.

3.2 PAIRED FRAGMENTS FROM LARGER INSTANCES

Before execution, we specified a constructed stress test on 256 fresh geometries in each of six cells: $n \in \{50, 100, 200\}$, uniform or clustered sampling. On each geometry, nearest-neighbor construction followed by at most 100 deterministic 2-opt moves provides a feasible tour. We cut it into six directed paths and exchange suffixes of two non-source paths. The source path, all component heads, and the set of tails remain fixed, while two head–tail associations change. Singleton paths are allowed; only the two swapped paths must have an internal cut. We enumerate all $5! = 120$ cyclic component orders in each forest and compute Equation (3) under equal pair weights.

Conflicting optimal actions occur in 20.7–29.7% of pairs (Table 1). Positive mean regret is not limited to the selected witness. The construction also exposes a limitation: swapping suffixes can produce a poor counterfactual forest and does not preserve a policy’s visitation distribution. These figures establish the reach of the interface ambiguity within a precisely specified family, not its prevalence during ordinary solving.

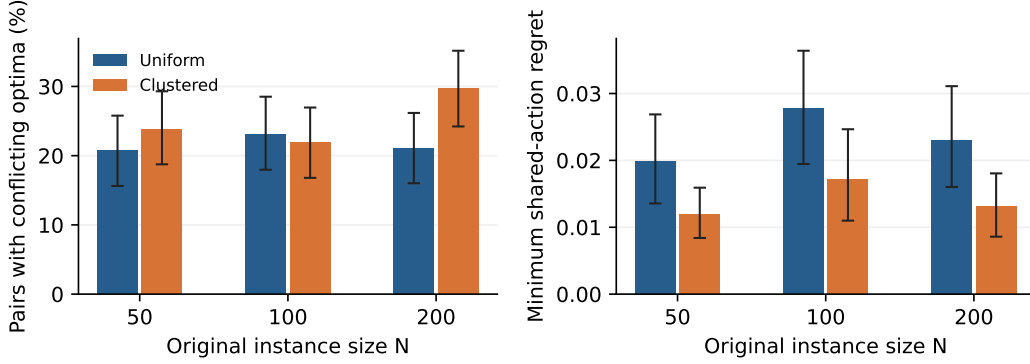


Figure 2: Constructed counterfactual pairs on larger instances, with six components per forest. Intervals bootstrap 256 independent geometries per cell, retaining both states within each unit. The rates describe this generator, not the occupancy of a deployed neural policy.

Distribution	N	Conflict (%)	Regret	95% CI
Uniform	50	20.7	0.0199	[0.0136, 0.0269]
Uniform	100	23.0	0.0278	[0.0195, 0.0364]
Uniform	200	21.1	0.0230	[0.0160, 0.0311]
Clustered	50	23.8	0.0119	[0.0084, 0.0159]
Clustered	100	21.9	0.0172	[0.0110, 0.0247]
Clustered	200	29.7	0.0131	[0.0086, 0.0181]

Table 1: Exact conditional observation regret on the larger-instance stress test. Costs use original Euclidean coordinates. Uniform and clustered instances have different distance scales, so absolute regrets are not a scale-invariant comparison. An unchanged-pair control has zero regret in all 1,536 cases.

3.3 LEARNING WITH AND WITHOUT CONNECTIVITY

We isolate connectivity in a supervised task with exact action values. Training uses 40,000 independent uniform nine-node geometries, each with two forests obtained from different matchings of three prescribed heads to three prescribed tails; the remaining three vertices are singletons, including the selected source. Validation has 3,000 different geometries. Each test condition has 4,500 new geometries from three data seeds. Both members of a pair always remain in the same split. Zero-shot tests vary the node count to 7, 11, and 13 or replace uniform coordinates by three Gaussian clusters.

Blind and aware predictors use the same three-layer, width-128 Transformer (415,489 parameters), with no positional encoding, and 10,000 AdamW updates of batch 512. Both receive coordinates relative to the source, head/tail roles, and singleton/non-singleton indicators. This enriched blind observation is deliberately more informative than the audited host interface. Only the aware predictor receives true partner-coordinate displacements. Models regress exact action regrets; validation regret selects the checkpoint. Three independent initialization seeds share the data and minibatch streams. An additional control trains the same network with independent random pairings, preserving active feature channels without providing true connectivity.

The true-pairing predictor reduces IID regret from 0.03422 to 0.00645 (81.2%). The paired absolute reduction is 0.02777, with a geometry-and-seed bootstrap 95% interval [0.02572, 0.02981]. The independently trained random-pairing control obtains 0.03374, close to the blind predictor. Thus active extra input channels alone do not explain the benefit. Table 2 retains distribution and size shifts and inexpensive nearest-head and cycle-cover assignment comparators. The assignment score uses a lower-bound relaxation; it is not an exact-completion oracle.

Enumerating all six matchings instead of a sampled pair yields an exact Bayes risk of 0.02831 for the enriched blind observation on uniform geometries. The aware model achieves 0.00633 on this same family (Figure 3). Revealing one prescribed partner reduces the risk to 0.01622; revealing

Test	Blind	Random pairing	True pairing	Assignment	Nearest
Uniform 9	0.03422	0.03374	0.00645	0.03629	0.07744
Clustered 9	0.01726	0.01778	0.00649	0.01667	0.03354
Uniform 7	0.03338	0.03502	0.00693	0.04154	0.08148
Uniform 11	0.03392	0.03389	0.01227	0.03045	0.07081
Uniform 13	0.03332	0.03295	0.01887	0.02626	0.06704

Table 2: Exact fixed-source completion regret, averaged over three training seeds and paired states. This is not complete-tour solver performance. The random-pairing control and assignment-relaxation heuristic are supplementary controls designed after pilot validation. All final selection uses validation. Every condition has 4,500 independent test geometries.

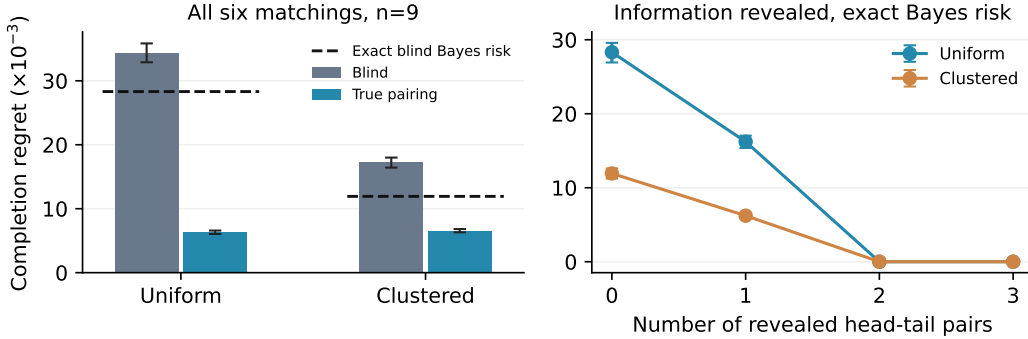


Figure 3: A supplementary exhaustive observation-class evaluation on the same held-out geometries. Left: all $3! = 6$ possible matchings, with learned-model means over three seeds and geometry-bootstrap intervals. Right: reveal the partners of a predetermined subset of heads and recompute exact Bayes risk. These curves concern the specified observation family, not native-host occupancy.

two determines the third and reduces it to zero. This full-family computation avoids confusing a pair-aware oracle with the Bayes limit for the actual controlled observation.

Swapping the two true associations at test time while keeping model weights fixed raises IID regret to 0.06271. This is a sensitivity intervention and a distribution change, not a retrained baseline. Together with the independently trained random-pairing control, it supports the role of connectivity in the conditional task. Independent Held-Karp checks of stored labels agree with enumeration to floating-point precision; coordinate hashes exclude leakage, blind predictions are exactly identical within each pair, and the largest measured permutation-equivariance error is 2.98×10^{-7} .

4 FROM CONDITIONAL DECISIONS TO COMPLETE TOURS

4.1 ARCHIVED ADAPTATION EVIDENCE

We separately audit an existing official-host experiment comprising AM and ICAM, route continuation, trained random-source forests, and learned-source forests, with three joint-training seeds per arm. Models receive 2,500 endpoint-supervision updates followed by 300 reinforcement-learning updates; the three seeds share the relevant warmup. The dataset, trajectory count, and updates are matched, but training and inference wall times are not. This is a limited adaptation screen, not evidence of convergence. We recompute its tables from archived per-instance costs and retain its negative outcomes.

Learned source selection beats trained random selection on uniform TSP200, yet costs remain 23.50% above route for AM and 6.26% above route for ICAM (Table 3). All three predefined size/distribution settings favor route in mean cost. This motivates diagnosis but does not show that aliasing caused the entire loss. Nor do 300 updates establish an upper limit on a method trained to convergence.

Host	Test	Route	Random	Learned	Learned excess (%)
AM	U200	11.5579	18.2188	14.2735	23.50
AM	C200	8.0337	17.8553	11.9997	49.37
AM	U500	19.7023	31.7857	24.5729	24.72
ICAM	U200	10.8307	12.6999	11.5086	6.26
ICAM	C200	5.2792	8.1126	6.4374	21.94
ICAM	U500	16.9091	21.1741	18.3969	8.80

Table 3: Archived adaptation screen, kept separate from new interventions. U200/C200/U500 denote uniform TSP200, clustered TSP200, and uniform TSP500. Entries average three joint seeds sharing warmup; lower is better. Learned excess is relative to adapted route. No inference-time equivalence or state-repair conclusion follows from this table.

Host	Training	Test	Native	Route-B	Route-A	Forest-B	Forest-A
ICAM	Frozen	U100	7.828	7.860	7.851	9.290	9.095
ICAM	Frozen	C100	4.012	3.938	3.917	5.579	5.536
ICAM	Frozen	U200	10.837	10.871	10.861	13.397	13.209
ICAM	Frozen	C200	5.256	5.168	5.150	8.175	8.272
AM	Frozen	U100	8.089	8.155	8.148	11.671	11.268
AM	Frozen	C100	4.727	3.782	3.791	5.331	5.125
AM	Frozen	U200	11.621	11.600	11.563	17.323	16.877
AM	Frozen	C200	7.147	5.011	4.990	8.153	7.781
ICAM	Decoder	U100	7.806	7.868	7.836	9.049	9.661
ICAM	Decoder	C100	4.074	3.946	3.959	5.143	4.731
ICAM	Decoder	U200	10.787	10.900	10.826	12.939	13.848
ICAM	Decoder	C200	5.284	5.184	5.194	7.147	6.435
AM	Decoder	U100	8.077	8.178	8.139	10.851	10.493
AM	Decoder	C100	4.812	3.762	3.758	4.925	4.698
AM	Decoder	U200	11.559	11.515	11.508	15.546	15.133
AM	Decoder	C200	7.269	4.980	4.958	6.943	6.620

Table 4: Complete-tour cost for official-backbone interventions (lower is better). B/A denote blind/aware component inputs. Forest uses a uniformly random open source; route continues the current path. Each learned entry is the mean of three training seeds. Frozen and decoder-adapted suites use different final-test seeds and are not paired across training phases.

Native deployment budget is a separate issue. In an additional archived evaluation, ICAM with all n starts and eight geometric transformations has gaps to a feasible LKH reference of 0.27% on U200, 0.76% on U500, and 10.62% on C200. These are not certified optimality gaps. They also cannot be compared as equal-compute results to a single forest rollout. A weak single-start baseline can exaggerate practical headroom.

4.2 CONTROLLED COMPONENT-STATE INTERVENTION

We next test whether the component-state intervention transfers to complete tours with official AM and ICAM backbones. For each backbone we compare four arms—route or uniformly random source selection, each with blind or aware component inputs—under three independent training seeds. The residual has 58,306 parameters in every arm. In the frozen setting only the residual is trained; in the decoder setting the encoder remains frozen while the native decoder and residual are adapted. Each arm receives 12,000 supervised updates on the same native-plus-2-opt teachers. Checkpoints are selected by complete tour validation before four disjoint final-test sets are generated. No learned source policy is used in this experiment.

The complete-tour results do not mirror the conditional diagnostic. With a frozen AM host, awareness lowers forest cost in all four cells, but aware forest remains 35–56% worse than aware route. Frozen ICAM effects are smaller and mixed across the four cells. When the ICAM decoder is also adapted, awareness reduces teacher-state NLL from 1.183 to 0.539 on U100 and from 1.662 to 0.588 on U200, yet complete-tour forest cost becomes 7.47% and 7.44% worse, respectively. It improves clustered C100 and C200 forest tours by 7.53% and 9.18%, but still trails aware route by 20–28%. Across all 16 host, training, and test cells in Table 4, aware forest is worse than aware route.

Three additional source-action RNG seeds reproduce the broad frozen-host conclusion and show that the sign of small individual ICAM effects is not stable. A target-tail shuffling sensitivity test confirms that fitted models use the supplied correspondence, but it does not construct an alternative legal forest and is not a causal estimate. These outcomes separate two claims: connectivity is valuable in exact conditional decisions, while this residual, teacher, and random-source training protocol does not turn that information into a competitive full-tour forest solver.

5 RELATED WORK

State abstractions preserve different levels of decision information (Li et al., 2006). BQ-NCO applies bisimulation quotienting to obtain reduced constructive problems with optimal-solution guarantees (Drakulic et al., 2023). Our use of conditional Bayes regret examines an information-losing adapter in the opposite direction: enlarging construction freedom while retaining an interface designed for a narrower state family. It is a diagnostic application of established principles.

Multi-fragment construction is an established routing heuristic (Mamano et al., 2019), and path contraction is also standard. DRHG explicitly represents a retained segment using both endpoint coordinates and learns reconnection (Li et al., 2025); this is a direct precedent for our connectivity intervention. L2C-Insert learns insertion positions (Luo et al., 2025), and Learning to Segment identifies stable segments to reduce repeated search (Ouyang et al., 2026). We do not claim that flexible construction, paired endpoints, or compression is novel, and do not present our limited adapters as stronger solvers than these systems.

Variable-order learning is also established in constraint satisfaction (Song et al., 2019). Order-invariant reinforcement learning randomizes variable order in black-box optimization (Goudet et al., 2026), while MACSIM uses multi-action assignments and set-based learning (Luttmann & Xie, 2026). Such training can address ordering and equivalent trajectories, but it does not automatically reveal omitted connectivity. Conversely, adding connectivity does not prove that a learned order is useful. These are separate hypotheses requiring separate controls.

6 DISCUSSION AND LIMITATIONS

The evidence supports testing observation sufficiency before attributing forest-decoder failures to the optimizer or the usefulness of flexible order. It does not establish a new group algorithm or a general superiority of multi-fragment routing. The central remaining question is whether measured conditional information loss predicts gains on the states a solver actually visits. Exact small-state supervision, constructed larger-state pairs, and limited pretrained-host adaptation each answer part of that question. They are not interchangeable measurements. Strong solver claims require direct modern baselines, trained ordering controls, independent end-to-end seeds, and matched inference-time frontiers.

AI USE STATEMENT

Generative AI tools assisted with the formulation and proof drafts, experimental design, code implementation and checking, literature review, result interpretation, and manuscript and figure preparation. The released artifacts distinguish executed experiments from analysis and preserve their outputs. This statement does not assert that independent human verification has already been completed.

REPRODUCIBILITY STATEMENT

The supplementary code provides deterministic data generators, exact-completion checks, numerical interval verification, training and evaluation scripts, per-instance outputs, configurations, and file hashes. Protocols distinguish archived evidence from new experiments and record budget amendments. Appendix sections give the assumptions, data generation, and statistical units needed to interpret these artifacts.

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A ASSUMPTIONS AND LIMITS OF THE REGRET STATEMENT

The observation-conditioned distribution in Proposition 1 is part of the problem specification. A constructed equal-weight pair yields a valid risk statement for that pair. If a decoder’s true observation class also contains additional states, then an oracle that is told which pair was sampled has extra information. Its average pairwise minimum can be smaller than the minimum achievable by a decoder lacking pair identity. We therefore label pairwise and full observation-class enumeration separately.

Randomization cannot exploit hidden state unless its random seed is itself correlated with that state conditional on the observation. Recurrent history, dynamic encoder inputs, or a source-selector hidden message violate the interface assumed by the bound. A learned source also changes both the conditional state distribution and the information conveyed by its selected index. None of our conditional lower bounds is an impossibility theorem for joint source–target policies.

If approximate action values satisfy $|\hat{Q}_F(s, a) - Q_F(s, a)| \leq \epsilon$ uniformly, the estimated regret gap differs from the exact gap by at most 2ϵ : each of the two terms in Equation (3) changes by at most ϵ . Feasible heuristic completions only upper-bound action values, and usually do not provide such two-sided error control. A disagreement between heuristic completions alone does not certify positive irreducible regret.

B CONTRACTION AND COMPLETION COUNTS

Every completion must retain all fixed directed path edges. Contract each path to a component with its ordered head and tail. A completed tour corresponds to a cyclic ordering of the m distinguishable components, with one connector from each component’s tail to the next component’s head. Conversely any cyclic component order expands into a Hamiltonian tour because the graph is complete. Fixing one component to remove rotational duplicates gives $(m - 1)!$ completions. Specifying a legal connector merges two components and leaves $(m - 2)!$ orders. With one component only its closure remains.

Any optimal completion assigns a successor to every open source, so fixing which source is queried first does not change the optimal completion value provided all its legal heads remain available. It can nevertheless change the outcome of a restricted neural policy. These statements concern the directed fixed-fragment problem. Reversing a fragment changes its fixed successor assignments; destroy-and-repair, sparse graphs, capacity constraints, and time windows require different transition and feasibility analyses.

C NUMERICALLY CERTIFIED WITNESS

The witness uses the following recorded decimal coordinates, with zero-based vertex numbering:

Vertex	x	y
0	0.9667947888374329	0.144645094871521
1	0.16519033908843994	0.6777374148368835
2	0.01400136947631836	0.032273709774017334
3	0.27172255516052246	0.45283347368240356
4	0.11830371618270874	0.4929446578025818
5	0.49647223949432373	0.7829722166061401
6	0.6254234313964844	0.34477776288986206

Each decimal is interpreted as an exact rational. For every squared distance q , integer square roots bound $\sqrt{q}$ between adjacent multiples of 10^{-15} . Summing the lower and upper edge bounds along all anchored tours and taking actionwise minima bounds each conditional optimal completion cost. For $L_s(a) \leq Q_s(a) \leq U_s(a)$, a valid lower bound on the equal-pair regret is

$$\frac{1}{2} \min_a \sum_s L_s(a) - \frac{1}{2} \sum_s \min_a U_s(a),$$

with the upper bound obtained by interchanging L and U . The verification uses only Python’s standard library and checks all 720 tours. This is a certificate for the specified decimal instance, distinct from claiming exact real arithmetic on unspecified binary-coordinate inputs.

D LARGER-INSTANCE STRESS GENERATOR

Uniform instances sample n coordinates independently in $[0, 1]^2$. Clustered instances sample four centers uniformly in $[0.15, 0.85]^2$, assign each vertex uniformly to a center, add independent Gaussian noise with standard deviation 0.04, and clip to the unit square. There are 256 geometries per size/distribution cell and no test-dependent selection. Seeds are $91972000 + 10n + b$, where $b = 0$ for uniform and $b = 1$ for clustered.

A nearest-neighbor tour begins at vertex zero. Deterministic best-improvement 2-opt accepts at most 100 improving moves. This is a feasible construction, not an optimality certificate or a benchmark competitor. Five cut positions are sampled without replacement among the $n - 1$ gaps in its linearized order, giving six paths. If fewer than two non-source paths have length at least two, cuts are resampled. From eligible paths, two are sampled and cut internally; their suffixes are exchanged. The source is the tail of path zero and is unchanged. All path heads and the set of tails remain unchanged.

For each pair, all 120 directed cycles on six components are enumerated. Q-values include fixed internal lengths, but these cancel separately within each forest when computing regret. Both forests are retained, including all zero-regret pairs. A control duplicating the original forest must yield zero. In floating-point summaries, a pair is marked conflicting when its minimum shared-action regret exceeds 10^{-9} . Bootstrap intervals use 4,000 resamples of whole geometries, never treating the two forests as independent observations. These sampling rules are deliberately explicit because different fragment and counterfactual generators can give different conflict rates.

E CONTROLLED NEURAL PREDICTION PROTOCOL

For $n = 2k + 3$ vertices, vertex zero is the selected singleton source; vertices $1, \dots, k$ are nontrivial component heads and vertices $k + 1, \dots, 2k$ are their tails. The last two vertices are singletons. An initial uniform random matching defines k length-two components; swapping two randomly selected matches constructs the second forest. Coordinates are sampled independently of the matching. The number of components is $k + 3$, so enumerating anchored component cycles is tractable for the tested $k = 2, 3, 4, 5$. Distances and action values are computed in float64 from coordinates stored as float32.

The shared node features are source-relative coordinates, a head indicator, a non-head indicator, a source indicator, and a non-singleton indicator. Two additional channels contain displacement to the paired endpoint in the aware condition and zeros in the blind condition. Both heads and tails receive the displacement. There are no positional embeddings. The network uses an eight-dimensional input projection to width 128, three pre-norm Transformer layers with eight heads and feed-forward width 256, GELU activation, and zero dropout. A shared normalized MLP outputs one scalar per candidate head. Mean-squared error to exact regret trains the scores; the chosen action minimizes the predicted score over legal heads.

AdamW uses learning rate 3×10^{-4} , weight decay 10^{-5} , gradient-norm clipping at one, and batch size 512 for 10,000 updates. Validation runs at the first update and every 500 updates thereafter; the lowest mean exact validation regret selects the checkpoint. The seeds are 19091941, 19091942, and 19091943. Train and validation geometry seeds are 190919101 and 190919201; each test condition concatenates 1,500 geometries from each of seeds 190919301–190919303. The clustered condition uses three centers in $[0.15, 0.85]^2$, Gaussian noise of standard deviation 0.06, and clipping to the unit square. All train, validation, and test geometries remain disjoint; reusing a data seed at another size does not create an identical geometry.

The independently trained random-pairing control resamples a uniform matching for every training example on every update using a separate RNG. It retains all roles and active feature channels but makes the displayed matching independent of the label-generating matching. Validation and test use fixed independent corruption streams. Test-time wrong pairing instead exchanges the two members’ true associations while keeping the aware model fixed. These two experiments answer different

questions. Both are supplementary mechanism controls designed after initial pilot validation; neither selects the main predictors using the final test.

The supplementary full-family evaluation holds geometry and roles fixed and enumerates all six matchings for $k = 3$. It therefore computes the exact Bayes regret for the enriched blind observation and the specified uniform matching prior. Progressive revelation groups the six matchings by the partners of heads $1, \dots, r$ and averages each group’s optimal shared action with its probability. With two known partners the third is determined. No learned predictor or test-based selection is needed for this calculation.

Statistical summaries first average both paired states within each geometry. Model means then average three training seeds. Geometry-only intervals condition on these trained models; additional crossed bootstrap intervals resample both whole geometries and the three seeds. The latter do not compensate for having only three training seeds. The assignment comparator forces each candidate first action and scores an assignment relaxation of the remaining directed component problem, without requiring a single cycle. Its lower-bound scores are used as a heuristic ranking, not as exact Q-values.

F ARCHIVED HOST STUDY DETAILS

The earlier study starts from the official AM TSP100 epoch-99 checkpoint and the official ICAM checkpoint. Each host supplies greedy tours for 2,048 uniform TSP200 training geometries, improved by up to 200 best-improvement 2-opt moves. Route and random-order endpoint warmup use 2,500 updates of batch 32 at learning rate 10^{-5} ; the selected warmup is shared across the three subsequent joint seeds 1234, 2468, and 4321. The joint stage has 300 updates with four independent instances and four stochastic trajectories per instance, leave-one-out REINFORCE baselines, host learning rate 10^{-5} , selector learning rate 10^{-4} , and gradient clipping at one.

Joint validation uses 64 instances every 50 updates, including the initial checkpoint. The primary test has 256 uniform TSP200 instances; secondary tests have 64 clustered TSP200 and 64 uniform TSP500 instances. The reported seed uncertainty is conditional on shared warmup. Random source actions use a documented fixed evaluation RNG; their variation across multiple action seeds was not estimated in that archived study. Both hosts were evaluated on the same predefined conditions; failing cells were retained.

The earlier prototype computes endpoint distributions for all sources at every step. Its forest rollouts are slower than native route construction. The results are matched by samples and updates, not GPU seconds. The original source selector uses detached endpoint-probability summaries; its gradient does not include derivatives through those summaries into the host. These details limit conclusions about efficiency and optimization.

G ARTIFACT ORGANIZATION AND RESEARCH BOUNDARIES

The root protocol distinguishes four work packages: exact controlled learning, official-host representation intervention, larger-instance counterfactual stress testing, and theory/literature verification. Analysis scripts regenerate tables and figures from machine-readable results. Historical cost vectors are identified by content hashes and are not relabeled as new measurements.

The manuscript provides no claim of a new group-theoretic pruning algorithm, no guarantee for arbitrary constrained routing, and no equal-time solver ranking. Full-source freedom and complete-tour reachability concern the action space, not a deterministic neural policy’s actual support. More training, a learned source policy, an alternative loss, and additional decoding starts are different interventions and must be compared explicitly.